

Medvale Internal Medicine

Chapter 1K: Vascular Medicine, Hypertension, Venous Thromboembolism, and Shock

Revised edition with expanded shock section

Why this chapter matters

Vascular disease is where cardiology becomes whole-body medicine. The same atherosclerotic, thrombotic, inflammatory, and hemodynamic processes that cause coronary disease also produce hypertensive emergency, aortic dissection, abdominal aortic aneurysm, peripheral artery disease, acute limb ischemia, venous thromboembolism, and shock. These disorders are high yield because they force the clinician to decide quickly whether the patient needs blood pressure reduction, anticoagulation, thrombolysis, surgery, fluids, vasopressors, or immediate source control.

This revised chapter keeps the vascular structure from the prior version but adds a fuller shock framework. Shock is not simply low blood pressure. Shock is end-organ hypoperfusion. A patient may be in shock because the pump has failed, the tank is empty, the pipes are pathologically dilated, or blood cannot physically move through the circulation. The practical bedside question is: what is the dominant mechanism, and what intervention reverses it fastest?

Core clinical pattern

Hypertensive emergency injures organs because pressure is too high. Shock injures organs because perfusion is too low. Aortic dissection and ruptured aneurysm can create both problems in sequence: severe hypertension may trigger vascular injury, then bleeding or malperfusion can produce shock.

Numbers to know

Topic	High-yield number or cutoff	Clinical meaning
Hypertensive emergency	SBP ≥ 180 and/or DBP ≥ 120 plus acute target-organ damage	Treat with IV agents and monitored reduction.
Hypertensive urgency	Severe BP elevation without acute target-organ damage	Lower gradually with oral agents, often over 24–48 hr.
Initial BP reduction in hypertensive emergency	Reduce mean arterial pressure by about 25% in 1–2 hr	Avoid sudden hypoperfusion of brain, heart, and kidneys.
Aortic dissection anti-impulse therapy	Target SBP < 120 mm Hg if tolerated and HR about 60–80 bpm	Reduce shear stress on the aortic wall.
AAA size for repair, classic threshold	> 5.5 cm in many patients, or symptomatic/rapidly expanding	Rupture risk begins to outweigh operative risk.
PAD ABI	Normal about 1.0–1.4; PAD < 0.90 ; noncompressible often > 1.3 –1.4	ABI converts symptoms into objective arterial disease.
Severe limb ischemia	ABI often < 0.40 when rest pain/tissue loss is present	Threatened limb; vascular evaluation.
Acute limb ischemia tissue window	Skeletal muscle tolerates roughly 6 hr of severe ischemia	Immediate heparin and urgent revascularization planning.

Septic shock fluid anchor	30 mL/kg crystalloid early in sepsis-induced hypoperfusion/septic shock	Initial resuscitation, then reassess.
Septic shock MAP goal	MAP ≥ 65 mm Hg in many adults	Vasopressors if hypotension persists after fluids.
Hypovolemic shock compensation	Compensation begins failing after about 20–25% blood volume loss	Class III–IV shock usually needs aggressive resuscitation.
Cardiogenic shock definition anchor	SBP < 90 with low urine output and elevated filling pressure in classic teaching	Pump failure; do not drown with fluids.
SIRS criteria	2 or more: T > 38 C or < 36 C, HR > 90 , RR > 20 or PaCO ₂ < 32 , WBC $> 12k$ or $< 4k$ or $> 10\%$ bands	Older sepsis screening framework, still exam-relevant.
Warfarin VTE target	INR 2.0–3.0	Traditional therapeutic range.

Hypertensive emergency and hypertensive urgency

Hypertensive emergency is severe blood pressure elevation with acute target-organ damage. The number alone is not enough. A patient with SBP 210 mm Hg and no symptoms, no renal injury, no neurologic deficit, no pulmonary edema, and no ischemia is not the same as a patient with SBP 185 mm Hg and aortic dissection. The emergency is defined by organ injury, not by the absolute blood pressure in isolation.

Classic teaching uses systolic blood pressure ≥ 180 mm Hg and/or diastolic blood pressure ≥ 120 mm Hg as the severe range that should trigger a search for acute organ injury. Target organs include the brain, retina, heart, aorta, kidneys, and lungs. Neurologic manifestations include altered mental status, severe headache, seizure, intracranial hemorrhage, hypertensive encephalopathy, and posterior reversible encephalopathy syndrome. Cardiac and pulmonary manifestations include acute coronary syndrome, acute heart failure, and pulmonary edema. Renal manifestations include acute kidney injury or oliguria. Aortic dissection is an especially important vascular complication because the treatment target is not simply lower pressure but lower aortic wall shear stress.

The first management principle is controlled reduction, not normalization. In most hypertensive emergencies, the mean arterial pressure is reduced by about 25% during the first 1–2 hours, followed by slower reduction over the next day. Rapid overcorrection can produce cerebral, coronary, or renal hypoperfusion because autoregulation has adapted to chronically elevated pressures. IV medications such as nicardipine, clevidipine, labetalol, esmolol, nitroprusside, or nitroglycerin are selected based on the clinical syndrome. Hypertensive urgency lacks acute target-organ damage and is usually treated with oral agents and close follow-up rather than ICU-level IV therapy.

Graph-RAG association

Hypertensive emergency $\rightarrow$ acute target-organ damage $\rightarrow$ IV therapy and monitored reduction. Hypertensive urgency $\rightarrow$ no acute target-organ damage $\rightarrow$ oral therapy and gradual reduction.

Aortic dissection

Aortic dissection occurs when blood enters the aortic media through an intimal tear and separates the layers of the vessel wall. Long-standing hypertension is the most common risk factor in many adult patients, but connective tissue disease, bicuspid aortic valve, coarctation, pregnancy, cocaine or stimulant use, trauma, and atherosclerotic

ulceration also increase risk. The classic pain is abrupt, severe, tearing, ripping, or stabbing chest or back pain, but the absence of classic words does not exclude the diagnosis.

Stanford classification is central. Type A dissection involves the ascending aorta and is a surgical emergency because it can extend into the coronary arteries, aortic valve, pericardium, or arch vessels, causing myocardial infarction, acute aortic regurgitation, tamponade, stroke, or death. Type B dissection is limited to the descending aorta distal to the left subclavian artery and is often treated medically unless complicated by rupture, malperfusion, uncontrolled pain, uncontrolled hypertension, or expansion.

Initial treatment is anti-impulse therapy. Beta-blockade is given first to reduce heart rate and the force of left ventricular ejection. If vasodilators are given before rate control, reflex tachycardia can increase shear stress and worsen the dissection. A modern target is SBP <120 mm Hg or the lowest pressure that maintains perfusion, with heart rate roughly 60–80 beats per minute. CT angiography is the usual rapid diagnostic test in stable patients. TEE is especially useful when the patient is too unstable to leave the bedside. Anticoagulation or fibrinolysis can be catastrophic if dissection is mistaken for ACS.

Dissection clue	Reason it matters
Tearing pain radiating to back	Classic but not always present.
Pulse deficit or BP differential	Branch vessel involvement.
Neurologic deficit	Carotid or spinal malperfusion.
New diastolic murmur	Acute aortic regurgitation from proximal dissection.
Hypotension/JVD	Possible tamponade from rupture into pericardium.
Type A	Ascending aorta; surgery.
Type B	Descending aorta; medical unless complicated.

Abdominal aortic aneurysm

An abdominal aortic aneurysm is an abnormal dilation of the abdominal aorta, most commonly infrarenal and associated with atherosclerotic degeneration of the wall. Risk increases with age, smoking, male sex, hypertension, family history, and connective tissue disease. Many AAAs are discovered incidentally, but symptomatic expansion or rupture produces abdominal, flank, back, groin, or leg pain. The classic triad of rupture is abdominal or back pain, hypotension, and a pulsatile abdominal mass, but many patients do not present with all three.

Ultrasound is the screening and sizing test of choice in stable patients because it is fast, noninvasive, and highly sensitive for aneurysm diameter. CT angiography is used when detailed anatomy is needed for operative planning or when rupture is suspected in a stable patient. Unstable patients with suspected rupture should not be delayed by unnecessary imaging when operative management is needed.

Repair is recommended for symptomatic aneurysms and for large aneurysms where rupture risk outweighs operative risk. A classic threshold is >5.5 cm in many patients, though individualized factors such as sex, growth rate, symptoms, anatomy, and operative risk matter. Rapid expansion, tenderness, pain, or rupture signs shift the decision toward urgent repair regardless of a simple size threshold.

Peripheral arterial disease

Peripheral arterial disease is usually atherosclerotic obstruction in lower-extremity arteries. It is strongly associated with smoking, diabetes, hypertension, hyperlipidemia, chronic kidney disease, and coexisting coronary or cerebrovascular disease. The symptom pattern is exertional muscle ischemia: reproducible cramping pain with walking that improves with rest. Calf claudication often reflects femoropopliteal disease, buttock or hip claudication suggests aortoiliac disease, and foot pain or ulcers suggest more distal disease.

The ankle-brachial index translates bedside suspicion into objective evidence. ABI is the ratio of ankle systolic pressure to brachial systolic pressure. Normal is roughly 1.0–1.4. PAD is usually ABI <0.90. Severe ischemia often produces very low ABI values, commonly <0.40 in rest pain or tissue loss. A falsely high ABI may occur in diabetes or chronic kidney disease because calcified arteries are noncompressible; toe-brachial index or vascular imaging may be needed.

Treatment begins with risk reduction: smoking cessation, antiplatelet therapy when indicated, statin therapy, blood pressure and diabetes management, foot care, and structured exercise therapy. Cilostazol can improve walking distance in selected patients but should be avoided in heart failure. Revascularization is considered for lifestyle-limiting claudication despite medical therapy and exercise, or for chronic limb-threatening ischemia with rest pain, nonhealing wounds, or gangrene.

PAD finding	Association
Reproducible calf pain with walking	Intermittent claudication.
Pain at rest, worse with elevation	Severe ischemia.
Pain relieved by dependency	Gravity improves perfusion.
Shiny skin, hair loss, thick nails	Chronic poor arterial supply.
Ulcers on toes/pressure points	Arterial ulcer pattern.
Pallor on elevation, rubor on dependency	Advanced PAD physiology.

Acute limb ischemia

Acute limb ischemia is a vascular emergency caused by sudden arterial obstruction. The most common embolic source is the heart, especially atrial fibrillation, recent MI with mural thrombus, prosthetic valves, endocarditis, or myxoma. In-situ thrombosis can occur on preexisting PAD or aneurysmal disease. The limb becomes threatened because muscle and nerve tissue tolerate only a limited period of severe ischemia; classic teaching uses about 6 hours as an important salvage window.

The classic six Ps are pain, pallor, pulselessness, poikilothermia or coldness, paresthesia, and paralysis. Pain and sensory symptoms may occur early; paralysis and anesthesia are late and ominous. The initial step is immediate systemic anticoagulation with IV unfractionated heparin unless contraindicated, followed by urgent vascular surgery or interventional evaluation. Imaging should define anatomy when it will not delay revascularization of a threatened limb. Embolism may be treated with embolectomy, catheter-directed therapy, or surgery; thrombosis on chronic disease may require thrombolysis, angioplasty, stenting, bypass, or hybrid approaches.

Six Ps of acute limb ischemia

Pain, pallor, pulselessness, poikilothermia, paresthesia, paralysis. Paralysis and profound sensory loss suggest a late threatened limb.

Cholesterol embolization and mycotic aneurysm

Cholesterol embolization occurs when cholesterol crystals shower from proximal atherosclerotic plaque, often after angiography, vascular surgery, thrombolytic therapy, or anticoagulation. It causes small-vessel occlusion rather than a large thrombus. The typical associations are blue or purple toes, livedo reticularis, renal dysfunction, abdominal pain or GI bleeding, eosinophilia, and intact pulses because the large arteries may remain open. Treatment is supportive, with risk-factor control and avoidance of unnecessary anticoagulation if there is no other indication.

A mycotic aneurysm is an infected aneurysm. The term does not require fungal infection; bacterial infection is common. Blood cultures are often positive, and treatment requires IV antibiotics plus surgical or endovascular management depending on anatomy and rupture risk.

Deep venous thrombosis and pulmonary embolism

Venous thromboembolism is driven by Virchow triad: venous stasis, endothelial injury, and hypercoagulability. Risk factors include age, immobilization, surgery, trauma, malignancy, pregnancy or estrogen therapy, inherited thrombophilia, prior DVT or PE, heart failure, and central venous catheters. DVT classically causes unilateral leg swelling, pain, tenderness, warmth, erythema, and sometimes a palpable cord, but the classic findings are neither sensitive nor specific. Many DVTs are subtle or asymptomatic, and many painful swollen legs are not DVT.

Diagnosis is based on pretest probability. In low or intermediate probability patients, a negative high-sensitivity D-dimer can help rule out DVT or PE. In high probability patients, imaging is required because the post-test risk after a negative D-dimer may remain unacceptable. Duplex ultrasonography is the usual first test for suspected lower-extremity DVT, especially proximal DVT. CT pulmonary angiography is commonly used for PE when imaging is needed and renal function and contrast risk allow.

Anticoagulation prevents clot propagation and embolization. DOACs are commonly used in many stable nonpregnant patients without contraindications. LMWH is often preferred in pregnancy and may be favored in some cancer-associated thrombosis scenarios depending on patient factors. Warfarin requires overlap with parenteral anticoagulation until INR is therapeutic, with a target INR of 2.0–3.0. Initial treatment duration is often at least 3 months, then extended therapy is individualized based on provoking factors, bleeding risk, recurrence risk, and patient preference.

VTE concept	High-yield association
DVT source of PE	Ileofemoral/pelvic veins have high embolic risk.
Massive PE	Hypotension or shock physiology.
Submassive PE	RV strain without sustained hypotension.
Thrombolysis	Consider in massive PE if no contraindication.
IVC filter	PE prevention when anticoagulation is contraindicated or fails; does not treat DVT.
Post-thrombotic syndrome	Chronic pain, swelling, venous insufficiency after DVT.
Phlegmasia cerulea dolens	Severe venous obstruction causing painful blue swollen limb.

Shock: general framework

Shock is a state of circulatory failure producing inadequate tissue oxygen delivery and end-organ hypoperfusion. Hypotension is common but not required at the earliest stage; patients may initially compensate with tachycardia and vasoconstriction. As shock progresses, organs fail. The most exam-relevant organ signals are altered mental status from cerebral hypoperfusion, oliguria from renal hypoperfusion, lactic acidosis from anaerobic metabolism, cool or warm skin depending on vascular tone, and respiratory distress from metabolic acidosis or pulmonary edema.

The source material divides shock into four major categories: hypovolemic, cardiogenic, distributive, and obstructive. Hypovolemic shock is an empty-tank problem. Cardiogenic shock is a pump-failure problem. Distributive shock is a vascular-tone problem with pathologic vasodilation and maldistributed flow. Obstructive shock is a circulation-blockage problem in which blood cannot fill or leave the heart normally because of tamponade, tension pneumothorax, or massive pulmonary embolism. These categories are not academic; they determine whether the patient needs fluids, inotropy, vasopressors, drainage, decompression, thrombolysis, or revascularization.

Shock type	CO	SVR	PCWP/filling pressure	Typical clue
Hypovolemic	Low	High	Low	Cool, clammy, flat neck veins.
Cardiogenic	Low	High	High	Cool, clammy, JVP and pulmonary edema often present.
Distributive	High early; may fall late	Low	Low/normal	Warm early in sepsis/anaphylaxis; neurogenic may have bradycardia.
Obstructive	Low	High	Variable, often elevated right-sided pressures	JVP often elevated; cause-specific signs.

Initial shock approach

Address airway, breathing, circulation. Obtain IV access, cardiac monitor, ECG, CBC, BMP, lactate, cultures if infection is suspected, bedside ultrasound/echo when cardiac or obstructive shock is possible, and start empiric treatment while identifying the cause.

Initial approach to a patient in shock

The first step is stabilization. Airway and breathing are assessed because severe shock can cause respiratory failure, aspiration risk, altered mental status, or severe metabolic acidosis requiring ventilatory support. Circulation is addressed with large-bore IV access or intraosseous access, blood pressure monitoring, frequent reassessment, and early consideration of arterial line or central venous access if vasopressors are needed. In undifferentiated shock, isotonic crystalloid is often started, but the clinician must reassess quickly because cardiogenic shock can worsen if excessive fluids are given.

The bedside exam should look for fever, warm extremities, rash, wheeze, urticaria, bleeding, vomiting, diarrhea, trauma, melena, hematemesis, chest pain, dyspnea, unilateral leg swelling, muffled heart sounds, JVD, asymmetric breath sounds, new murmur, abdominal tenderness, and neurologic deficits. Point-of-care ultrasound is especially useful because it can rapidly identify poor LV function, RV dilation, pericardial effusion, tamponade physiology, pneumothorax, IVC size, pulmonary edema, free intraperitoneal fluid, and abdominal aortic aneurysm.

If hypotension persists despite appropriate fluid resuscitation, vasopressors are used. Norepinephrine is the usual first-line vasopressor for septic shock because it raises vascular tone with less tachyarrhythmia than dopamine. Vasopressin is often added when norepinephrine requirements rise. Epinephrine can be added in refractory shock. Dobutamine is considered when low cardiac output persists despite adequate perfusion pressure, especially in cardiogenic shock or septic myocardial dysfunction.

Cardiogenic shock

Cardiogenic shock occurs when the heart cannot generate enough cardiac output to maintain tissue perfusion. Acute MI is the most common cause in many clinical settings, but decompensated heart failure, arrhythmias, myocarditis, valvular catastrophe, ventricular septal rupture, papillary muscle rupture, and cardiac arrest can all cause cardiogenic

shock. Classic teaching defines cardiogenic shock as persistent hypotension, often SBP <90 mm Hg, with evidence of hypoperfusion such as urine output <20 mL/hr, and elevated left ventricular filling pressure.

The hemodynamic pattern is low cardiac output, high SVR from compensatory vasoconstriction, and elevated PCWP if left-sided filling pressures are high. The exam often shows cool clammy extremities, altered mental status, oliguria, narrow pulse pressure, pulmonary congestion, and elevated JVP. A new murmur after MI suggests mechanical complication. Pulmonary edema with hypotension should raise concern for pump failure rather than simple volume overload.

Treatment begins with ABCs and correction of the cause. MI requires urgent revascularization when appropriate. Tamponade requires pericardiocentesis or surgical drainage. Severe valvular lesions may require surgical or catheter-based intervention. If volume overloaded, loop diuretics, nitrates, dialysis, or ultrafiltration may be used carefully, but hypotension limits afterload reduction. Inotropes such as dobutamine or milrinone can improve contractility. Norepinephrine is often used when hypotension is severe. Mechanical circulatory support, such as intra-aortic balloon pump, Impella, VA-ECMO, or surgical ventricular assist devices, may be considered in selected patients as a bridge to recovery, revascularization, transplant, durable device, or decision.

Intra-aortic balloon pump concept

The balloon inflates during diastole to augment coronary perfusion and deflates before systole to reduce afterload. It is a support device, not definitive treatment for the underlying cause.

Hypovolemic shock

Hypovolemic shock is decreased circulating volume causing reduced venous return, reduced preload, reduced stroke volume, and reduced cardiac output. Hemorrhage is the most urgent form, but vomiting, diarrhea, burns, dehydration, diuresis, and third-spacing can also produce hypovolemia. The body initially compensates with tachycardia, vasoconstriction, increased contractility, and venous tone. Once blood loss exceeds roughly 20–25% of circulating volume, compensatory mechanisms begin to fail and hypotension becomes more likely.

Class	Blood volume loss	Pulse	BP	Capillary refill / urine / CNS
I	10–15%	Normal	Normal	Usually normal.
II	20–30%	>100	Normal or narrowed pulse pressure	Anxious, urine output falls.
III	30–40%	>120	Decreased	Confused, delayed refill, urine about 20 mL/hr.
IV	>40%	>140	Markedly decreased	Lethargic/comatose, negligible urine.

Treatment is volume restoration and source control. In hemorrhagic shock, apply direct pressure when possible, activate massive transfusion protocols when indicated, and pursue surgical, endoscopic, interventional radiology, or obstetric control of bleeding. Balanced resuscitation with packed red blood cells, plasma, and platelets is used for severe hemorrhage. In nonhemorrhagic hypovolemia, isotonic crystalloid is usually appropriate, followed by electrolyte correction and treatment of the cause. Urine output is a practical marker of resuscitation response, but mental status, lactate clearance, skin perfusion, blood pressure, and bedside ultrasound are also useful.

Septic shock and distributive shock

Septic shock is a form of distributive shock caused by infection-driven inflammation, vasodilation, capillary leak, and cellular/metabolic dysfunction. Early septic shock often has low SVR and high or normal cardiac output, producing warm flushed skin and bounding pulses. As myocardial depression, capillary leak, hypovolemia, and organ failure progress, the patient can become cool and mottled. Common sources include pneumonia, urinary tract infection, intra-abdominal infection, cholangitis, meningitis, soft-tissue infection, catheter infection, and endocarditis.

Management is simultaneous resuscitation and source control. Blood cultures are ideally obtained before antibiotics if this does not delay therapy. Broad-spectrum IV antibiotics are started rapidly and narrowed when culture data return. Initial fluid resuscitation for sepsis-induced hypoperfusion or septic shock commonly uses at least 30 mL/kg crystalloid within the first 3 hours, with frequent reassessment. If hypotension persists, norepinephrine is the preferred first-line vasopressor in many adults, with MAP target around 65 mm Hg. Vasopressin is commonly added when norepinephrine dose requirements are rising. Source control may require drainage of abscess, removal of infected lines, debridement, surgery, or biliary/urinary decompression.

Other distributive shocks share low SVR but differ in cause. Anaphylactic shock results from mast-cell mediator release with vasodilation, capillary leak, bronchospasm, urticaria, and airway edema; treatment centers on intramuscular epinephrine, airway management, fluids, antihistamines, bronchodilators, and corticosteroids as adjuncts. Neurogenic shock results from loss of sympathetic tone after spinal cord injury, severe head injury, spinal anesthesia, or pharmacologic blockade. It classically causes hypotension with bradycardia and warm skin because vascular tone is lost without the expected tachycardic compensation.

SIRS and sepsis exam anchor

SIRS is 2 or more of: temperature $>38^{\circ}\text{C}$ or $<36^{\circ}\text{C}$, HR >90 , RR >20 or $\text{PaCO}_2 <32$, WBC $>12,000$ or $<4,000$ or $>10\%$ bands. Sepsis is infection with organ dysfunction; septic shock is persistent hypotension requiring vasopressors and/or hypoperfusion despite fluids in modern frameworks.

Obstructive shock

Obstructive shock occurs when cardiac output is limited by a mechanical block to filling or outflow. The big three are cardiac tamponade, tension pneumothorax, and massive pulmonary embolism. The hemodynamic theme is low cardiac output with compensatory high SVR. JVP is often elevated because venous return is blocked or right-sided pressures are high, but the rest of the presentation depends on the cause.

Cardiac tamponade causes impaired diastolic filling from pericardial pressure. The key associations are hypotension, elevated JVP, muffled heart sounds, pulsus paradoxus, tachycardia, and echo evidence of right-sided chamber collapse. Treatment is pericardiocentesis or surgical drainage when unstable. Tension pneumothorax causes obstructive shock by increasing intrathoracic pressure, collapsing venous return, shifting mediastinal structures, and compressing the great veins. It is a clinical diagnosis in an unstable patient and requires immediate needle decompression followed by chest tube. Massive PE blocks pulmonary arterial flow and acutely increases RV afterload, causing RV dilation, septal bowing, hypotension, syncope, hypoxemia, and shock; treatment may require systemic thrombolysis, catheter-directed therapy, or surgical embolectomy in addition to anticoagulation.

Obstructive cause	Key clue	Definitive treatment direction
Tamponade	JVD, hypotension, muffled sounds, pulsus paradoxus, pericardial effusion	Pericardiocentesis or surgical window.
Tension pneumothorax	Hypotension, unilateral absent breath sounds, tracheal deviation late	Needle decompression then chest tube.

Massive PE	Hypotension, hypoxemia, RV strain, risk factors for VTE	Anticoagulation plus thrombolysis/thrombectomy if unstable.
------------	---	---

Integrated shock decision map

A useful mental model is to ask four questions. First, is the tank empty? If there is bleeding, dehydration, GI loss, burns, or flat neck veins with small ventricles, think hypovolemia and give volume while controlling losses. Second, has the pump failed? If there is MI, severe HF, pulmonary edema, new murmur, poor LV function, or high filling pressures, think cardiogenic shock and avoid reflexive large-volume fluids. Third, are the pipes dilated? If fever, infection, anaphylaxis, spinal cord injury, warm skin, or low SVR features dominate, think distributive shock and use fluids plus vasopressors, epinephrine for anaphylaxis, or spinal/neurogenic support. Fourth, is circulation blocked? If JVD is high with clear lungs, unilateral breath sounds, pericardial effusion, RV strain, or sudden dyspnea/syncope, think obstructive shock and remove the obstruction.

Bedside pattern shortcuts

Cold + wet = cardiogenic until proven otherwise. Cold + dry = hypovolemic or obstructive. Warm + hypotensive = early distributive shock, especially septic/anaphylactic/neurogenic. JVD is most characteristic of cardiogenic and obstructive shock.

Practice questions

Question 1

A 67-year-old man presents with BP 220/130, confusion, papilledema, and acute kidney injury. What is the diagnosis and initial strategy?

Answer: Hypertensive emergency. Begin IV antihypertensive therapy with monitored reduction, generally aiming to reduce MAP by about 25% in the first 1–2 hours rather than normalizing immediately.

Question 2

A 58-year-old woman has abrupt tearing chest pain radiating to the back. BP is different between arms. CTA shows ascending aortic dissection. What is the treatment category?

Answer: Stanford type A dissection. Give immediate anti-impulse therapy with beta-blockade and urgent surgical management.

Question 3

A 72-year-old smoker has calf pain after walking the same distance that improves with rest. ABI is 0.72. What is the diagnosis?

Answer: Peripheral arterial disease with intermittent claudication. Treatment begins with smoking cessation, statin/risk reduction, antiplatelet therapy when indicated, and structured exercise therapy.

Question 4

A patient with atrial fibrillation develops sudden severe leg pain, pallor, coldness, paresthesia, and absent Doppler pulses. What is the immediate treatment?

Answer: Acute limb ischemia. Start IV unfractionated heparin unless contraindicated and obtain urgent vascular surgery/interventional evaluation for revascularization.

Question 5

A 64-year-old with pneumonia remains hypotensive after fluids and has lactate elevation. What shock type is most likely and what vasopressor is typically first-line?

Answer: Septic distributive shock. Norepinephrine is typically the first-line vasopressor after appropriate fluid resuscitation.

Question 6

A patient after MI is hypotensive, cool, oliguric, has pulmonary edema, and bedside echo shows poor LV systolic function. What shock type is present?

Answer: Cardiogenic shock. Treat the underlying MI or mechanical complication, support perfusion with vasopressors/inotropes as needed, and avoid indiscriminate large-volume fluid loading.

Question 7

A trauma patient has hypotension, JVD, muffled heart sounds, and echo shows a large pericardial effusion with right atrial collapse. What is the treatment?

Answer: Obstructive shock from cardiac tamponade. Perform urgent pericardiocentesis or surgical drainage depending on context and stability.

Chapter summary

This chapter integrates vascular emergencies, venous thromboembolism, and shock. Hypertensive emergency is severe BP elevation with acute target-organ damage and requires IV monitored reduction; urgency lacks acute injury and is treated more gradually. Aortic dissection requires anti-impulse therapy and classification into type A surgical versus type B medical or complicated pathways. AAA risk rises with size and symptoms, with classic elective repair around >5.5 cm in many patients. PAD is diagnosed with ABI and treated with risk reduction, exercise, medications, and selective revascularization. Acute limb ischemia is recognized by the six Ps and requires immediate heparin plus urgent revascularization planning. VTE diagnosis depends on pretest probability, D-dimer, and imaging, while treatment centers on anticoagulation unless massive PE requires reperfusion therapy. Shock is end-organ hypoperfusion and is classified as hypovolemic, cardiogenic, distributive, or obstructive. The correct treatment depends on the mechanism: fill the tank, support the pump, tighten the vessels, or remove the obstruction.

Source and originality note

This chapter is original educational writing created for Medvale. The uploaded cardiology vascular and shock excerpts were used as coverage checklists for topic selection, numeric anchors, and exam-style associations. Current

guideline sources were used to cross-check selected modern management framing, especially PAD, aortic disease, and sepsis/shock resuscitation. No copyrighted figures or source prose are reproduced.